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Changepoint Detection for Net Electricity Demand Modelling in Great Britain

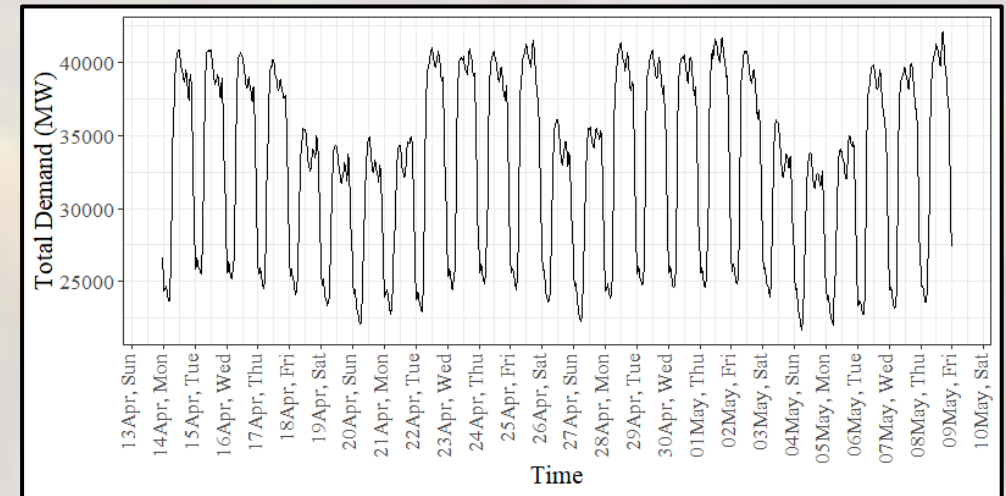
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24th January 2025

Scientific Challenge

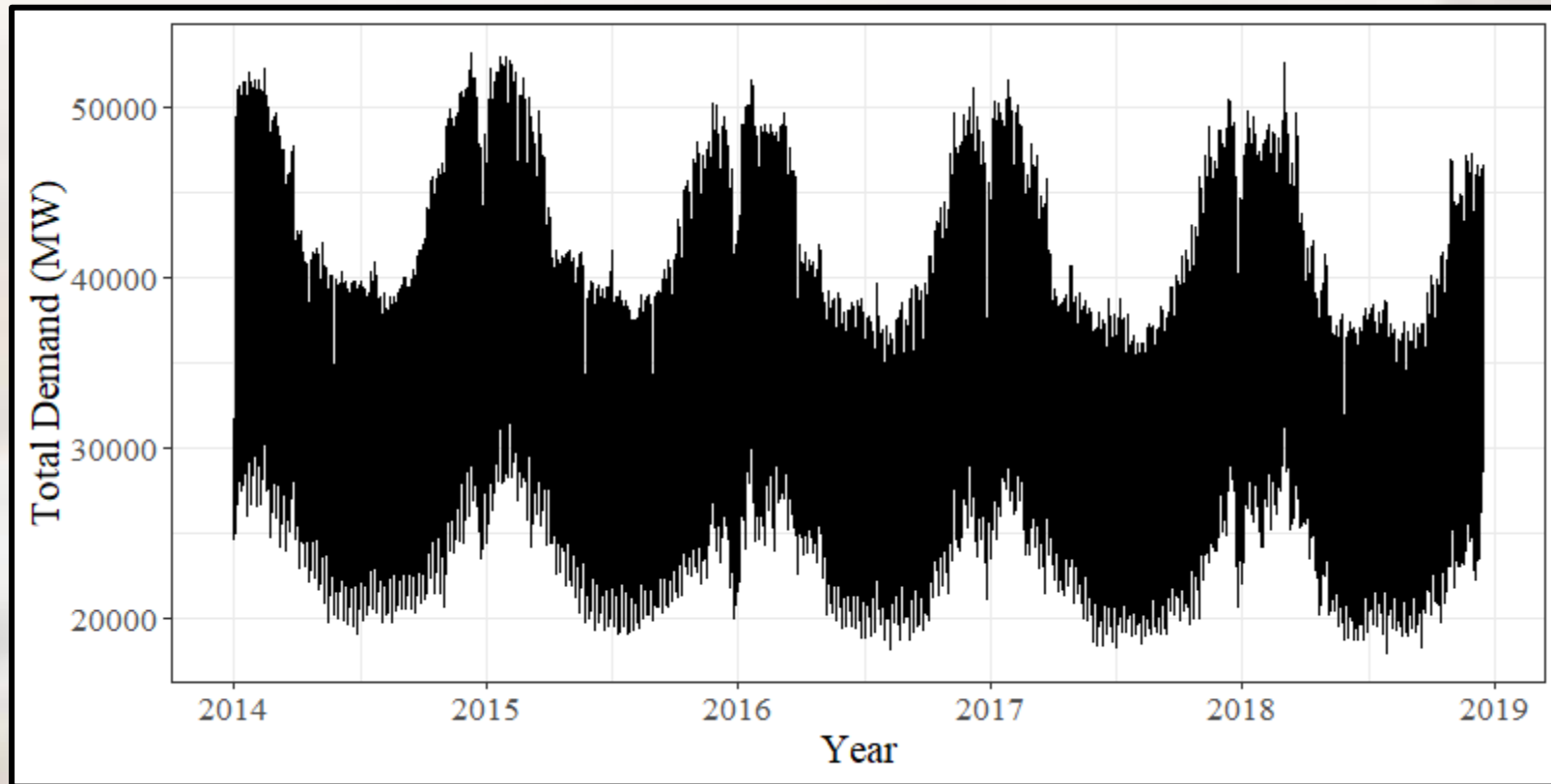
How can we model and forecast **electricity demand** accurately amidst increasing variability from **renewable energy integration**?

Key Challenges:

- Daily, weekly, and seasonal variations.
- Impact of renewable energy - solar and wind.
- Need to identify abrupt changes (changepoints) in demand patterns for better grid management.



Electricity Demand in Context

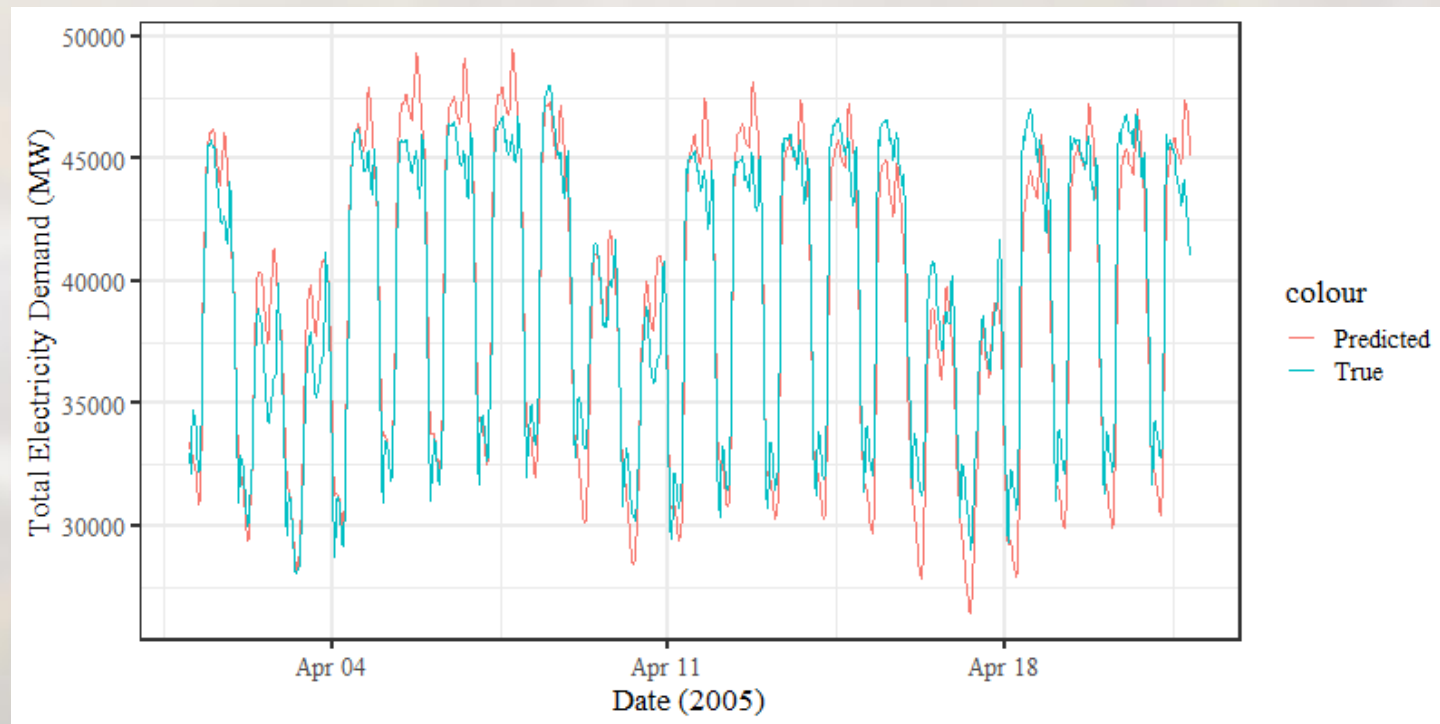


Aims of the Research

- Develop a comprehensive understanding of electricity demand patterns on a national scale (*Chapter 2*).
- Design a **changepoint detection algorithm** tailored to electricity demand (*Chapter 3*).
- Test and validate the algorithm using simulated and regional demand data (*Chapters 4 and 5*).

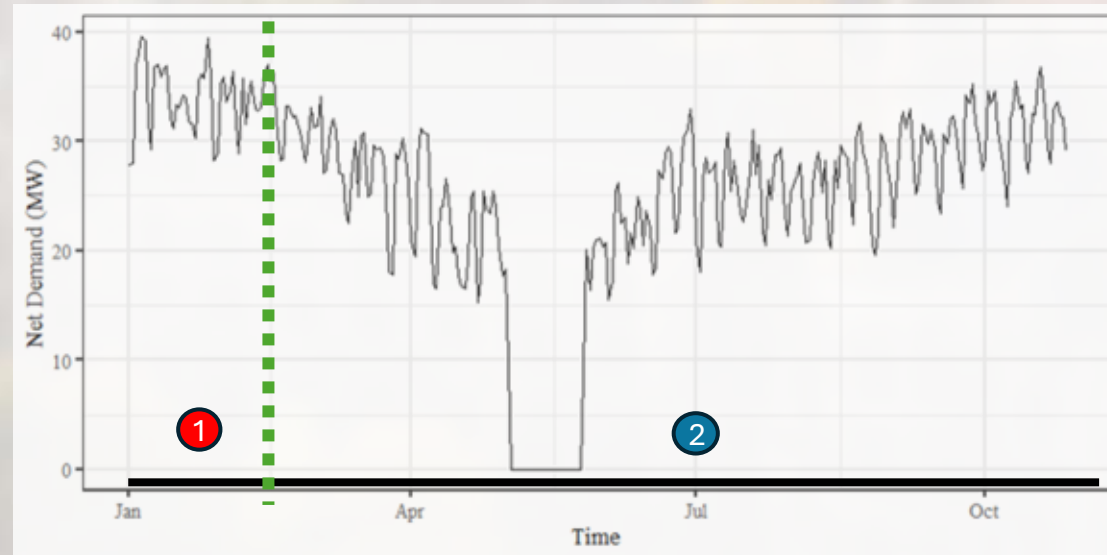
National Demand Modelling

- Overview of Generalised Additive Models (GAMs):
 - Captures non-linear trends and seasonal patterns.
 - Flexible and adaptable without extensive manual adjustment.



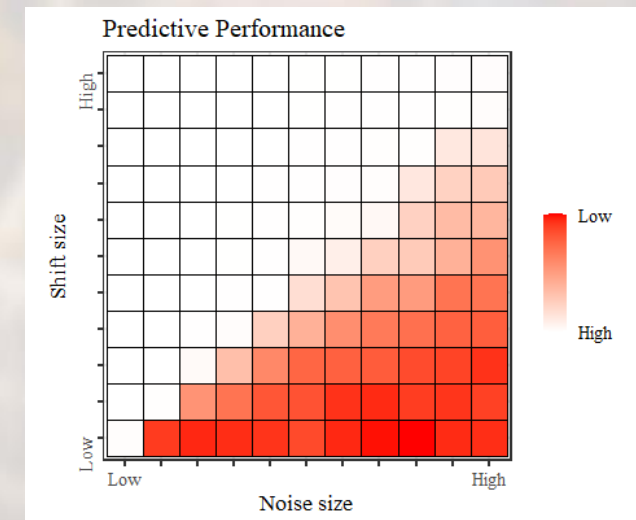
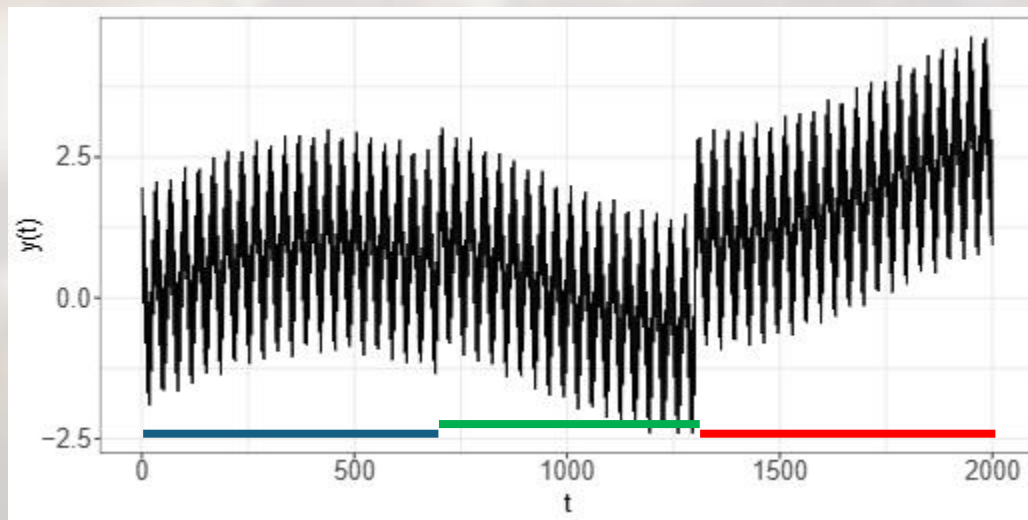
Changepoint Detection Algorithm

- Standard Method Evaluated: Binary Segmentation.
- Challenges:
 - Computational constraints with model-based cost functions.
- Novel Approach:
 - Custom changepoint detection algorithm integrated with GAMs.



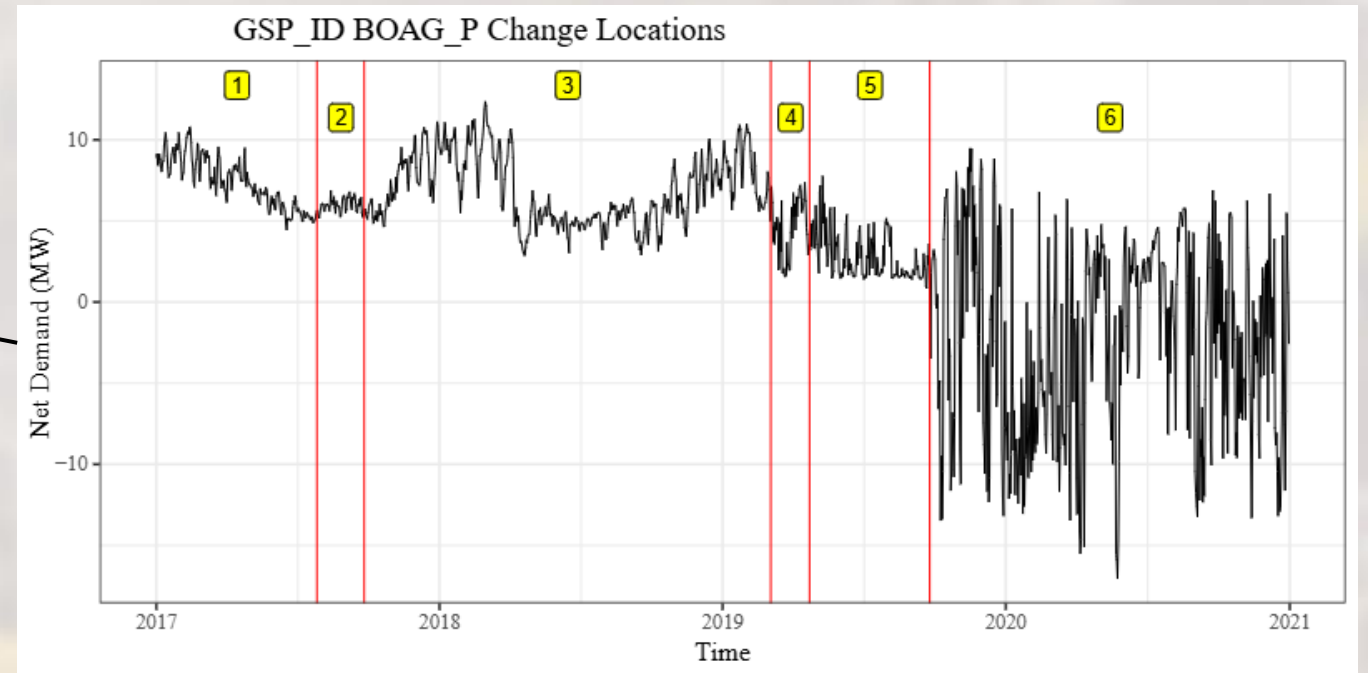
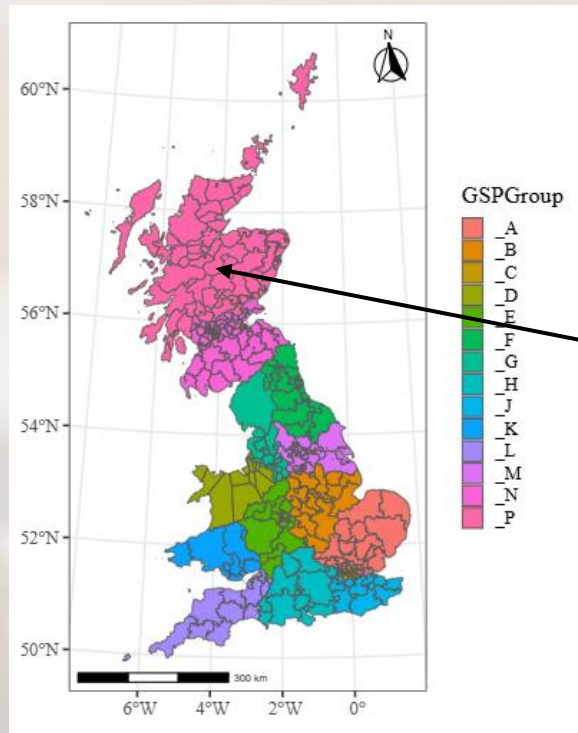
Simulation Study

- Purpose:
 - Test the algorithm under controlled conditions (Chapter 4).
- Key Findings:
 - Performance influenced by mean shifts and noise levels.
 - Effective at distinguishing true changepoints even with complex patterns.



Regional Applications

- Data:
 - Grid Supply Point (GSP) demand across regions.



Key Findings

- GAMs enhance model flexibility by accounting for non-linear trends and seasonality.
- Custom changepoint detection integrates effectively for electricity demand applications.
- Limitations:
 - Computational constraints.
 - Scalability challenges with large datasets.



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Thank you for your attention!